



SolMate

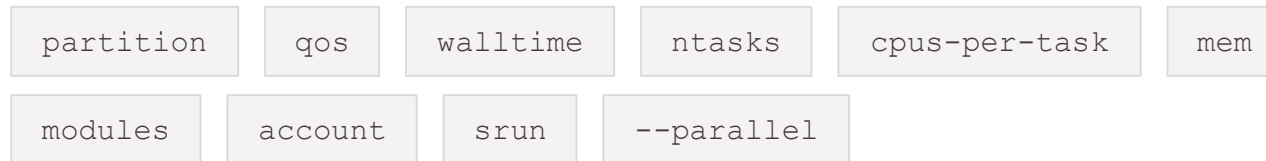
Powered by ASU AIR



A new researcher's first job is not a science problem.

They know the experiment. They have never written a scheduler request — and nothing in the error message teaches them how.

WHAT THEY ARE EXPECTED TO ALREADY KNOW



Today they learn it by submitting a job and reading the error.

That is a slow and discouraging way to learn — and it is how most people start.

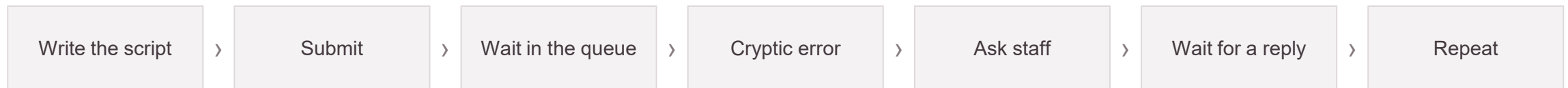
The barrier to doing research is not the science.

It is the syntax standing in front of it.



Finding out you were wrong should not take a day.

TODAY



Hours to days — and it costs someone else's time

WITH SOLMATE



Before the job ever reaches the queue

444 ms

to a scheduler recommendation

3.2 s

from failure evidence to a confirmed cause

Every loop closed in the browser is a loop not closed in the queue. That is the difference between finishing this week and next.



Research Computing keeps office hours. Jobs fail at 3 a.m.

A researcher who hits a wall at midnight should not lose a day waiting for a reply. SolMate answers immediately, with evidence, every hour of the week.

24 / 7

Available the moment a job fails, not the next business day

Instant

Feedback while the researcher is still thinking about the problem

Grounded

Every answer cites the evidence or the ASU rule behind it

The researcher stops guessing, and starts knowing. That confidence is the product.



AIR interprets. Deterministic code decides.

The model reads intent. Code owns every decision about what is safe to run.

01

DESCRIBE

YOU

Plain language. No scheduler vocabulary required.

02

VALIDATE

CODE

Every value checked against dated ASU partition, QoS and hardware limits.

03

RENDER

CODE

Only a complete, validated spec becomes a script. The model never emits shell text.

04

DIAGNOSE

AIR

A cause proposed by AIR, with every cited line verified against what was supplied.

No credentials. No submissions. No stored data. SolMate builds the commands; the researcher reviews and runs every one of them.



Describe the work. In plain language.

The screenshot shows the SolMate interface with two main panels. The left panel is titled '01 Describe the workload' and contains a sub-section 'Research workload' with a text input area. The text entered is: '/scratch/asurite/solmate-demo and the training script is train.py. I want to use the public partition and public QoS. I require 1 node, 1 task, 8 CPUs per task, 1 GPU, 32 GB of memory, and a 2 hour walltime. I need to train for 10 epochs with a batch size of 64.' Below the text area is a red button labeled 'Analyze with AIR'. The right panel is titled '03 Reviewed output' and contains the text 'Awaiting validated requirements'.

01 Describe the workload
AIR interprets the workflow as you pause, then identifies conflicts and missing decisions.

1 Describe **2 Confirm** **3 Review**

Research workload

/scratch/asurite/solmate-demo and the training script is train.py.

I want to use the public partition and public QoS. I require 1 node, 1 task, 8 CPUs per task, 1 GPU, 32 GB of memory, and a 2 hour walltime. I need to train for 10 epochs with a batch size of 64.

Analyze with AIR

Waiting for a pause...

03 Reviewed output
Complete the guided intake to generate a script.

Awaiting validated requirements

What the researcher types

```
Run a PyTorch ResNet-18  
training job on Sol.
```

What SolMate extracts

Every fact must quote the description word for word. Anything the researcher did not say stays empty.

No scheduler vocabulary required to start.



It asks. It does not guess.

AIR INTERPRETATION

The researcher wants to run a PyTorch ResNet-18 image classification training job on Sol with 10 epochs and a batch size of 64.

Detected software or framework: PyTorch

The request specifies 10000 GPUs which is highly unusual for typical ResNet-18 training, suggesting either a typo or misunderstanding of GPU requirements; the rest of the parameters are standard for ML training.

ASU RC DOCS

[Partitions and QoS](#) [Slurm SBATCH Job Scripts](#) [Job Statistics](#) [Python Environment Example](#)

AIR CORRECTED 10000 GPU -> 1 GPU

AIR ASKS

What is the expected dataset size and complexity for this ResNet-18 training?

Your answer

Add the detail you know; AIR will revise the plan.

Send answer to AIR

Conflicts to resolve

- Requested 10000 GPUs which is highly unusual for typical ResNet-18 training.

Other details that would improve the plan

- Is there a specific reason for requesting 10000 GPUs?
- What is the dataset size and complexity?
- Are there any specific data preprocessing steps required?
- Is there a preference for using Mamba or Conda environments?
- What is the expected training time with 1 GPU?
- Are there any specific libraries or dependencies required beyond standard PyTorch?

10,000 GPUs

Flagged as a conflict with the reason named, not silently accepted.

A blank stays blank

A general chatbot fills a missing value with a plausible number. SolMate names what is missing and waits for the researcher to answer.



A validated spec becomes a script.

TOTAL CORES	CORE-HOURS	MEMORY/TASK
8	16	32 GB
GPU-HOURS	PARTITION	QOS
2	public	public

Script

Explain

Check

Run

Generated Slurm script

Copy

```
#!/bin/bash
#SBATCH --job-name=resnet-demo
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=02:00:00
#SBATCH --partition=public
#SBATCH --qos=public
#SBATCH --output=/scratch/asurite/solmate-demo/resnet-demo-%x-%j.out
#SBATCH --error=/scratch/asurite/solmate-demo/resnet-demo-%x-%j.err
#SBATCH --chdir=/scratch/asurite/solmate-demo
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --gres=gpu:1

set -euo pipefail

srun python train.py --epochs 10 --batch-size 64
```

Every value confirmable

AIR SUGGESTIONS

Confirm all

☒ outputPath

/scratch/asurite/solmate-demo/resnet-demo...

LOW

▼ Why

AIR selected this editable value from the supplied workload and scheduler profiles. Assumes: This remains editable and requires researcher confirmation.. Measure next: Review this value before generating the script.

☒ errorPath

/scratch/asurite/solmate-demo/resnet-demo...

LOW

▼ Why

AIR selected this editable value from the supplied workload and scheduler profiles. Assumes: This remains editable and requires researcher confirmation.. Measure next: Review this value before generating the script.

☒ executable

python

LOW

▼ Why

AIR selected this editable value from the supplied workload and scheduler profiles. Assumes: This remains editable and requires researcher confirmation.. Measure next: Review this value before generating the script.

☒ args

train.py, --epochs, 10, --batch-size, 64

LOW

▼ Why

AIR selected this editable value from the supplied workload and scheduler profiles. Assumes: This remains editable and requires researcher confirmation.. Measure next: Review this value before generating the script.

Independent resource review: No GPU requirement specified in workload description despite suggesting 1 GPU. Workload is CPU-bound PyTorch training, not GPU-accelerated. Scheduler profile 'sol-public-class' supports 4 GPUs but workload does not require them.

The model never writes shell text. Code renders it, only from a complete spec.



A failed job, explained.

AIR suggests
Check available software and modules before installing another

[Related guide](#)

01

Supply failure evidence

Exact lines and metadata keep the diagnosis auditable.

Evidence	Deterministic checks	AIR finding
Cluster Sol		
▶ Collect evidence from Sol or Phoenix		
Slurm script <pre>#!/usr/bin/env bash</pre> Use generated script Load documented demo		
Job log or command output Paste the sbatch error, stderr, myjobs, seff, or sacct output here.		
Scheduler state	Exit code	

02

Evidence-based finding

No diagnosis yet.

Awaiting job evidence

Three answers, not one

CONFIRMED

Every cited line exists in what was supplied.

PROBABLE

Likely, with alternatives and a next step named.

INCONCLUSIVE

A queued job is not a failed job. SolMate stops.

“Inconclusive” is a feature, not a gap.



Grounded in ASU's own documentation.

A general chat bot guesses from the internet. SolMate reads the rules ASU actually publishes — and dates every one of them.

docs.rc.asu.edu

Every cluster rule cited to an ASU Research Computing page, with the date it was checked

9 profiles

Dated partition and QoS pairs across Sol and Phoenix. A pair not in that file cannot render

Real hardware

128-core Sol standard nodes, 48-core GPU nodes, and the Phoenix equivalents

This is why it works for a new researcher. The advice is not generic HPC folklore — it is ASU's partitions, ASU's QoS, ASU's nodes, on the day it was checked.



Nothing leaves ASU. Nothing sensitive is kept.

Research context is sensitive. We designed for that before we designed for features.

■ Inference stays on ASU AIR

Workload descriptions and logs never reach a commercial endpoint.

■ No credentials, ever

SolMate never logs in to Sol and never submits a job on anyone's behalf.

■ No database

Descriptions, scripts and logs are never persisted. An optional history keeps at most 12 sanitized outcome records in the researcher's own browser — resource numbers only, no text.

■ Redacted before the model sees it

Job IDs, account names, hosts, user paths, emails and keys are stripped from the payload.



Built to be handed over.

WEBSITE



<https://github.com/Saumya2402/solmate-asu-air/tree/continued-development>

Setup, the full workflow, the dated rule sources, and the safety boundaries — written for a researcher who has never used a cluster, not for us.

A tool that lowers a barrier has to be documented like one.

FUTURE EDITIONS

■ Ingest the documentation

Turn any center's published limits into a dated profile set, reviewed by a human before it goes live.

■ More research clusters

Phoenix already ships. Other public research centers become a profile set, not a rewrite.

■ Kubernetes, not only Slurm

The same intake and validation loop, rendering a different target.

■ Lab and course templates

A new student inherits a working baseline instead of a blank form.



SolMate turns cluster access from a syntax puzzle into a guided, auditable workflow.

Powered by ASU AIR and designed for ASU Research Computing.

1

Approve the concept

A guarded path from research intent to a reviewed submission, grounded in dated ASU rules.

2

Validate the Sol workflow

Walk the generated handoff with us and confirm the dry-run path end to end on a real account.

3

Test it with real workloads

Fixtures from actual failed jobs are the fastest way to make the diagnosis worth trusting.



Let researchers spend their time on research.

SolMate

ASU AIR Spark Challenge

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